Tensor computations

Equality of ranks for generic odd-order Hankel tensors

Difficulty: challenging **Importance:** interesting to the community

Statement

Fix an odd integer $m \geq 5$ and $n \geq 2$. A complex Hankel tensor H of order m and dimension n has entries

$$H_{i_1 \dots i_m} = h_{i_1 + \dots + i_m - m}, \qquad 1 \leq i_j \leq n,$$

for $h \in \mathbb{C}^{m(n-1)+1}$. Is there a nonempty Zariski-open subset of this Hankel tensor space on which

$$R(H) = R_{\mathrm{sym}}(H) = \underline{R}(H) = \underline{R}_{\mathrm{sym}}(H) = R_V(H)?$$

Here R is the minimum number of arbitrary complex rank-one tensor summands; R_{sym} restricts summands to complex multiples of $v^{\otimes m}$. Each underlined border rank is the least integer r admitting a sequence of complex tensors of the corresponding rank at most r converging entrywise to H. R_V further restricts v to

$$v(a,b) = (a^{n-1}, a^{n-2}b, \dots, b^{n-1}), \quad (a,b) \neq (0,0).$$

Thus limits defining symmetric border rank stay in the symmetric tensor space; limits defining ordinary border rank need not. Generically $R_V = \lceil (m(n-1)+1)/2 \rceil$.

Relevance

Hankel tensor decompositions encode sums of exponentials used in signal reconstruction. The conjecture compares structured decomposition lengths with the best unconstrained low-rank descriptions.

References

1. J. Nie and K. Ye, *Hankel Tensor Decompositions and Ranks*, SIAM J. Matrix Anal. Appl. 40 (2019). DOI; author preprint, §6, Question 6.1 and Conjecture 6.2, p.16; Corollary 3.3 gives the generic Vandermonde rank.
2. L. Qi, *Hankel Tensors: Associated Hankel Matrices and Vandermonde Decomposition*, 2014. Primary preprint, §1, equation (1), and §4, Theorem 3, for the structured decomposition background.

Status check — 2026-09-08

Reference 1's latest arXiv revision, January 28, 2019, explicitly poses this conjecture after the even-order and third-order results. Searches *"Hankel" "Conjecture 6.2"*, *"Hankel tensor" "rank" "odd order" conjecture*, and *"Hankel" "rank" "conjecture" "2026"* found no resolution. Status is supported by the originating paper and a bounded later search, without a fresh 2026 reaffirmation located.